

Ilaria Fontana

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PROFESSIONAL EXPERIENCE

Postdoctoral Research Associate, University of Arizona
Department of Mathematics

08/2024–present
Tucson, AZ (USA)

- Mentor: Lise-Marie Imbert-Gerard
- Development and analysis of quasi-Trefftz method for problems governed by vector-valued PDEs.
- Co-organizer of a minisymposium at CSE25
- Instructor for one section of First Semester Calculus

Visiting Assistant Professor, Northwestern University
Department of Engineering Science and Applied Mathematics (ESAM)

09/2022–08/2024
Evanston, IL (USA)

- Collaboration with the main developers of the **Dedalus project**: Daniel Lecoanet, Geoff Vasil, Keaton Burns.
- Analysis of different instances of the tau spectral method. In particular, investigation of the qualitative and quantitative properties of the approximate spectrum of two eigenvalues problems with fluid dynamics applications.
- Instructor for six classes of Multivariate Integral Calculus for engineering students.

Ph.D. Research Engineer, EDF (main French electric company)
R&D Division Lab Paris-Saclay

12/2018–02/2022
Palaiseau (France)

- Development of a posteriori error estimation applied to mechanical problems modeled with PDEs, and introduction of a fully adaptive algorithm including stopping criteria for the nonlinear solver and automatic tuning of some parameters. Implementation with **FreeFem++**.
- Development of models for the materials and their cohesive zones, both from a theoretical point of view, and with the creation of tailored FEA tools for the industry. Implementation with Python, Fortran and the numerical simulation tool **code_aster**.
- Collaboration between different divisions (also outside R&D) in order to provide customer-focused solutions to the final users. This taught me how to be autonomous, organized and willing to interact and work closely with diverse audiences.

EDUCATION

Ph.D. degree, University of Montpellier
Mathématiques et Modélisation (Mathematics and Modélisation)

12/2018–03/2022
Palaiseau (France)

- Collaboration with EDF (main French electric company) with **CIFRE agreement**
- Thesis title: *Interface problems for dam modeling (Problèmes d'interfaces pour les ouvrages hydrauliques)*
- Topics: Finite element approximation methods for PDEs, Modeling of hydraulic structure joints with cohesive zone model, A posteriori error estimate techniques for contact problems via equilibrated stress reconstruction
- Advisor: Daniele A. Di Pietro (University of Montpellier)
- Industrial supervisor: Kyrlo Kazymyrenko (EDF Lab Paris-Saclay)
- Research units: **IMAG - Institut Montpellierain Alexander Grothendieck**, and **IMSIA - Institute of Mechanical Sciences and Industrial Application**

Master degree, University of Udine
Applied Mathematics

10/2016–10/2018
Udine (Italy)

- Thesis title: *Numerical Bifurcation of Equations with Infinite Delay via Pseudospectral Collocation*
- Topics: DDEs with infinite delay, Dynamical systems generated by DDEs, Orthogonal polynomials
- Supervisor: Rossana Vermiglio
- Final grade: 110/110 cum laude

Bachelor degree, University of Udine
Mathematics

09/2013–10/2016
Udine (Italy)

- Thesis title: *Il teorema del passo di montagna (Mountain Pass Theorem)*
- Supervisor: Rodica Toader
- Final grade: 110/110 cum laude

TEACHING EXPERIENCES

MATH/DATA 363: Introduction to Statistical Methods, University of Arizona

2025

- Fall 2025: Instructor for one section - co-teaching with Joseph Watkins

MATH 122B: Calculus I, University of Arizona

2024

- Fall 2024: Instructor for one section

MATH 228-2: Multiple Integration and Vector Calculus, Northwestern University

2023–2024

- Winter 2024: Instructor for two sections - collaboration with Petia Vlahovska
- Fall 2023: Instructor for one section - collaboration with Daniel Lecoanet
- Spring 2023: Instructor for one section
- Winter 2023: Instructor for two sections - collaboration with Michael Miksis and Petia Vlahovska

PUBLICATIONS

Articles and Conference Papers

- I. Fontana, G. Bacquaert, D. A. Di Pietro, K. Kazymyrenko, *Hyperelastic nature of the Hoek–Brown criterion*. European Journal of Mechanics - A/Solids, 2026, **115**:105782.
DOI: [10.1016/j.euromechsol.2025.105782](https://doi.org/10.1016/j.euromechsol.2025.105782)
- [\[arXiv\]](#), [\[Hal\]](#)
- I. Fontana, L.-M. Imbert-Gérard, *Generalized Plane Wave quasi-Trefftz spaces for wave propagation in inhomogeneous media*. Accepted in Forum Acusticum/Euronoise 2025, the 11th EAA Annual European Conference on Acoustics and Noise Control Engineering, 2025.
- [\[arXiv\]](#)
- I. Fontana, D. A. Di Pietro, *An a posteriori error analysis based on equilibrated stresses for finite element approximations of frictional contact*. Computer Methods in Applied Mechanics and Engineering, 2024, **425**:116950. DOI: [10.1016/j.cma.2024.116950](https://doi.org/10.1016/j.cma.2024.116950)
- [\[arXiv\]](#), [\[Hal\]](#)

- D. A. Di Pietro, I. Fontana, K. Kazymyrenko, *A posteriori error estimates via equilibrated stress reconstructions for contact problems approximated by Nitsche's method*. Computers & Mathematics with Applications, 2022, **111**:61–80. DOI: [10.1016/j.camwa.2022.02.008](https://doi.org/10.1016/j.camwa.2022.02.008)
- [\[arXiv\]](#), [\[Hal\]](#)
- I. Fontana, K. Kazymyrenko, F. Escoffier, *Physics of interfaces: mechanical coupling of plasticity and damage*.
- In preparation. Submission to a journal preview in 2025.
- K. Burns, I. Fontana, D. Lecoanet, G. Vasil, *Spectrum analysis for the tau spectral method*.
- In preparation. Submission to a journal preview in 2025.

Other Ongoing Projects

- I. Fontana, L.M. Imbert-Gérard, Project on *Quasi-Trefftz methods for vectors-valued partial differential equations*
- A. Crippa, D. A. Di Pietro, I. Fontana, Project on *Adaptivity for fracture propagation*

Thesis

- I. Fontana, *Interface Problems for Dam Modeling (Problèmes d'interface pour les ouvrages hydrauliques)*, Ph.D Thesis. [Hal these](#)

CONFERENCES TALKS & SEMINARS

CSE25, SIAM Conference on Computational Science and Engineering March 2025
Quasi-Trefftz Methods with Applications to Maxwell's Equations Fort Worth, TX (USA)
 Abstract selected for oral presentation

Analysis, Dynamics, and Applications Seminar in the Department of Mathematics at the University of Arizona November 2024
Analysis of the Tau Method for Eigenvalue Problems Tucson, AZ (USA)
 Invited seminar

Analysis, Dynamics, and Applications Seminar in the Department of Mathematics at the University of Arizona November 2024
Analysis of the Tau Method for Eigenvalue Problems Tucson, AZ (USA)
 Invited seminar

Modeling and Computation Seminar in the Department of Mathematics at the University of Arizona September 2024
A Posteriori Error Analysis for Contact Problems with Application to Dam Modeling Tucson, AZ (USA)
 Invited seminar

WCCM-PANAC 2024, 16th World Congress on Computational Mechanics and 4th Pan American Congress on Computational Mechanics July 2024
Adaptive mesh algorithm for frictional contact problems based on a posteriori error analysis Vancouver (Canada)
 Invited seminar

- Spring 2024 Finite Element Circus at Brown University** April 2024
An equilibrated a posteriori error analysis for frictional contact problems Providence, RI (USA)
 Volunteer presentation
- Scintillae - Talk Alumni at the Scuola Superiore Universitaria Di Toppo Wassermann** December 2023
Metodi numerici per problemi meccanici (Numerical methods for mechanical problems) Udine (Italy)
 Invited talk by the Alumni Association
- ICCCM 2023, 7th International Conference on Computational Contact Mechanics** July 2023
Application of a posteriori error analysis to contact problems Turin (Italy)
 Abstract selected for oral presentation
- Spring 2023 Finite Element Circus at Bridgewater State University** March 2023
Dam modeling: Application of a posteriori error analysis to contact problems Bridgewater, MA (USA)
 Volunteer presentation
- Seminar in the Department of Mathematics at Purdue University** December 2022
Dam modeling: Application of a posteriori error analysis to contact problems West Lafayette, IN (USA)
 Invited seminar
- EFEF2020, 18th European Finite Element Fair** September 2021
A Posteriori Error Estimation via Equilibrated Stress Reconstruction for Unilateral Contact Problems Paris (France)
 Abstract selected for oral presentation
- USNCCM16, 16th U.S. National Congress on Computational Mechanics** July 2021
A Posteriori Error Estimation via Equilibrated Stress Reconstruction for Unilateral Contact Problems Online
 Abstract selected for oral presentation
- ADMOS 2021, 10th International Conference on Adaptive Modeling and Simulation** June 2021
A Posteriori Error Estimation via Equilibrated Stress Reconstruction for Unilateral Contact Problems Online
 Abstract selected for oral presentation
- Seminar in the CDLab of the University of Udine** June 2021
A Posteriori Error Estimation via Equilibrated Stress Reconstruction for Unilateral Contact Problems Online
 Invited seminar
- Congrès Français de Mécanique** August 2019
Lois mécaniques pour les ouvrages hydrauliques (Behavioral laws for hydraulic structures) Brest (France)
 Abstract selected for oral presentation
- Seminar in the CDLab of the University of Udine** October 2018
Numerical Bifurcation of Equations with Infinite Delay via Pseudospectral Collocation Udine (Italy)
 Master student seminar

Seminar in the CDLab of the University of Udine
Laguerre-type orthogonal polynomials and transformation of the Chebyshev nodes
Master student seminar

September 2018
Udine (Italy)

Seminar in the CDLab of the University of Udine
A model describing vapour and droplets
Master student seminar

July 2018
Udine (Italy)

Conferences Posters

Journées scientifiques du GdR MaNu
A posteriori error estimates via equilibrated stress reconstructions for contact problems approximated by Nitsche's method

October 2021
Le Croisic (France)

Colloque National MECAMAT 2019 – Rupture des Matériaux et des Structures
A model describing vapour and droplets

January 2019
Aussois (France)

GRANTS & SCHOLARSHIPS

CIFRE convention for Ph.D. funding
Industrial Agreements for Training through Research

2018–2022

Thesis Research fellowship

2018

Three months of Master thesis research at the Department of Mathematics and Statistics of the University of Helsinki (Finland) in collaboration with prof. Mats Gyllenberg and with the Biomathematics research group.

“Scuola Superiore” (School for Advanced Studies) fellowship

2013–2018

The *School for Advanced Studies* of the University of Udine is an institution for higher education besides the university. Admitted students follow additional courses and are examined every year on two occasions. The school covers university student fees and provides a residence for five years (Bachelor and Master degrees).

OUTREACH & SERVICE EXPERIENCES

Minisymposium Co-organizer, CSE25

March 2025

Co-organizer with Lise-Marie Imbert-Gérard of the minisymposium “Advances in Discontinuous Galerkin and Trefftz Methods” at the SIAM Conference on Computational Science and Engineering (CSE25).

WISE Mentor, University of Arizona

2024–current

Mentor for a Computer Science master student in the WISE mentorship program of the University of Arizona.

Math Postdoctoral Vice President, University of Arizona

2024–current

Vice President in the governance of postdocs in the Department of Mathematics of the University of Arizona.

Student tutor, University of Udine

2017–2018

Student tutor for Bachelor and Master degree in Mathematics of the University of Udine. In charge of teaching assistance and outreach activities.

- Effective communicator, advertising university programs in person and on social media to diverse audiences.
- Improving the customer experience, specifically in the Mathematics Department, by identifying bottlenecks in the course load and by proposing solutions to the department directors.

Student representative, University of Udine

2014–2018

Student representative for Bachelor and Master degree in Mathematics of the University of Udine.

TECHNICAL SKILLS

Programming languages

Python, C/C++, FreeFem++, Dedalus, Mathematica, MATLAB, Fortran, SQL, R

Other

L^AT_EX, Beamer, code_aster, Linux

LANGUAGES

Italian

Mother tongue

English

Workplace - Daily

French

Workplace - Daily

Spanish

High-school

Portuguese

Basic